

YASH BAVADIYA

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EDUCATION

Indian Institute of Technology, Jodhpur

Aug 2024 – Jul 2028

Bachelor's Degree - Computer Science and Engineering - CGPA - 7.21

Jodhpur, Rajasthan

TECHNICAL SKILLS

Languages: C++, C, Python, JavaScript, TypeScript, SQL

Core CS: Data Structures and Algorithms, Object-Oriented Programming, Software Design, Operating Systems, Database Management Systems, Computer Networks, Distributed Systems

AI Engineering: Prompt Engineering, Retrieval-Augmented Generation, Agentic Pipelines, LLM Evaluation, Context Management, FAISS, Hugging Face, PyTorch

Technologies/Frameworks: Node.js, Express, React, Next.js, Qt (C++), Prisma, PostgreSQL, MySQL, MongoDB, Redis, WebSockets, Docker, Linux/UNIX, REST APIs, TCP/IP

Practices & Tools: Git, CI/CD (GitHub Actions), Automated Testing, Agile/Scrum, Code Review, AI Coding Assistants (Claude Code, GitHub Copilot), Nginx, Postman

EXPERIENCE

Google Summer of Code 2026

May 2026 – Aug 2026

Kdenlive @KDE Community | Open Source Contributor, C++ and Qt

Remote

- Selected for **GSoC 2026** among 1,200 global contributors to design and ship Kdenlive's **RGB curve, gradient, and speed ramp editors** in C++ and Qt, owning features end to end from design through code review to release.
- Shipped **20 contributions** across Kdenlive and upstream repositories, resolving an **11-year-old bug** with a gap-detection algorithm now running for **41,000 monthly installs**, reviewed and merged by core maintainers.

Undergraduate Researcher

Jul 2025 – Nov 2025

Indian Institute of Technology, Jodhpur | Supervisor: Dr. Pratik Mazumder

IIT Jodhpur

- Engineered an offline **Retrieval-Augmented Generation pipeline** in Python, integrating **FAISS vector search** with 4-bit quantized Phi-3 and Flan-T5-XL under strict context-window and memory constraints.
- Benchmarked **prompt engineering** and retrieval strategies across **8 ROUGE metrics** (+**71.77%** over baseline), reaching **3 second query latency** on 300-page PDFs for **15 users** across 3 feedback-driven iterations.

PROJECTS

DaemonDoc (team of 2) | [GitHub](#) | [Live](#) | TypeScript, LLM APIs, Express, MongoDB, Redis

Jan 2026 – May 2026

- Built an **agentic LLM pipeline** on a **Redis (BullMQ)** job queue with webhook idempotency and loop guards, sustaining an **85% success rate** across **959 production runs** over 68 repositories for **50 users**.
- Cut per-run API cost roughly **10x** through **context management**, routing file paths and token estimates, never file content, to a lightweight model to select context, with failover across 6 provider keys.
- Owned the **GitHub integration and prompt-construction layers**, plus a Razorpay billing system secured with **HMAC-SHA256** webhook signature verification.

Flock | [GitHub](#) | [npm](#) | TypeScript, WebSocket, Redis, React, Docker

Feb 2026 – Jun 2026

- Released 3 npm packages enabling multiplayer cursors and live presence via a **10-line integration**, installable as a **55MB** self-hosted Docker image or npx.
- Implemented a horizontally scalable **Redis pub/sub relay** across server instances with keyspace-notification-driven eviction, reducing stale presence state to under **30 seconds** on ungraceful disconnects.
- Delivered cursor smoothing with mid-animation target updates, exponential backoff reconnection, and **58 automated tests** covering throttling, time-to-live eviction, and message delivery.

Photon | [GitHub](#) | C++20, Linux, UDP Multicast, Lock-Free Concurrency

Jun 2026 – Jul 2026

- Built a **NASDAQ ITCH 5.0 feed handler and matching engine**: zero-copy parsing, busy-pollled UDP multicast, and a **lock-free SPSC ring**, reaching **38ns median parse** and **452ns** socket-to-consumed latency via **rdtsdp** on a full day of real market data.
- Sustained **89k messages/sec** across **826 concurrent price-time priority order books** with zero hot-path heap allocation; verified by **27 GoogleTest** cases, **ASan/UBSan** presets, and clang-tidy CI.

COURSEWORK

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|----------------------------------|-------------------------------|------------------------------|
| • Data Structures and Algorithms | • Database Management Systems | • Pattern Recognition |
| • Object-Oriented Programming | • Operating Systems | • Probability and Statistics |
| • Software Engineering | • Machine Learning | • Discrete Mathematics |

ACHIEVEMENTS

- Solved **450+** algorithmic problems on **LeetCode** in C++; Max Rating **1429** on Codeforces across rated contests.
- JEE Advanced 2024: AIR **3449** of 180,000+ candidates; JEE Main 2024: **99.65** percentile, AIR **5680**.
- Merged **85+** open-source contributions across GitHub and GitLab, shipping reviewed fixes to **Deno**, **Adobe React Spectrum**, **Kubernetes SIGs**, **nlohmann/json**, **Vitest**, and **Kdenlive**.

LEADERSHIP

- **Junior Core Team Member**, DevUp Labs, IIT Jodhpur; mentored 30 students and ran 2 open-source workshops.
- **Core Member**, Programming Club and E-Cell, IIT Jodhpur; co-organized 5 competitive programming contests, 10 sessions, and 3 entrepreneurship events.